

CoRoT-2b

R-1788

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_130623 · created 2026-08-03T00:23:59+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456466.8253 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.211 +/- 0.01
Transit depth (Rp/Rs)^2	4.47 +/- 0.43 [%]
Orbital Inclination (inc)	87.53 +/- 1.29
Airmass coefficient 1 (a1)	1.018 +/- 0.025
Airmass coefficient 2 (a2)	-0.013 +/- 0.039
Scatter in the residuals of the lightcurve fit is	2.46 %
Best Comparison Star	#3 - [152, 180]
Optimal Method	PSF photometry
Transit Duration (day)	0.0973 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.211 ± 0.01 vs 0.1667 (deviation 4.43σ)
Duration fit vs published	0.0973 d vs 0.095 d (deviation 0.68σ)
Mid-time O-C	-6.2 min
Depth SNR	21.1
Residual scatter	2.46 %

Light curve

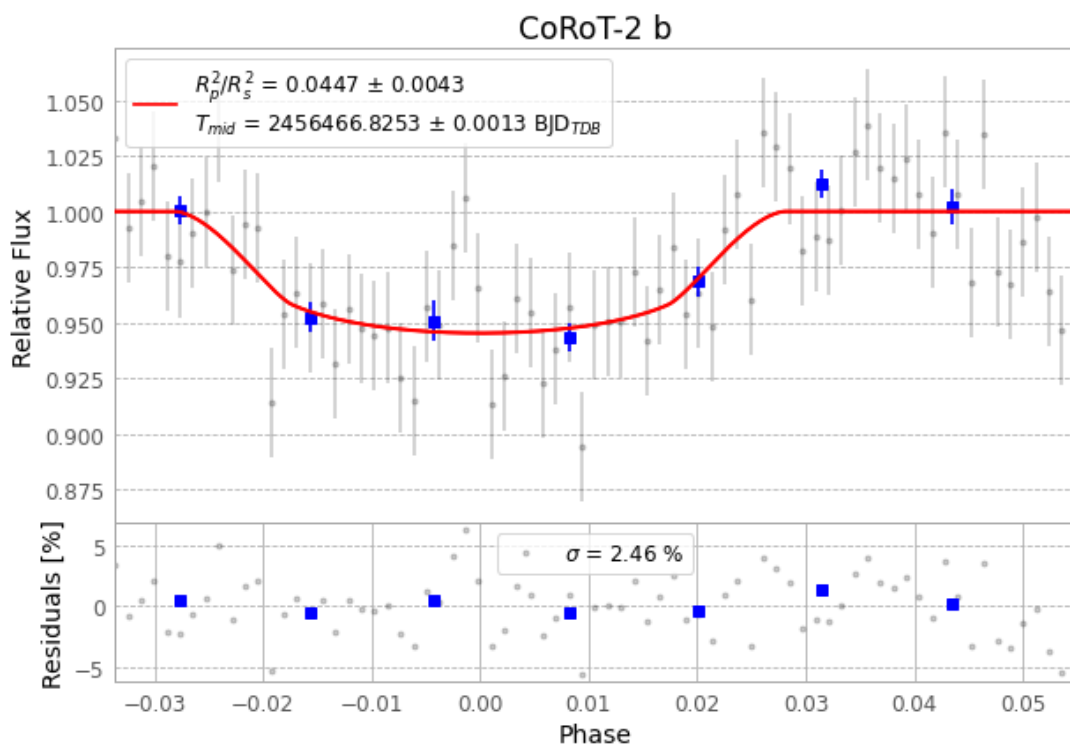


Figure 1 — FinalLightCurve_CoRoT-2 b_2013-06-22.png (SHA-256 39eb07bc2182b27d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [287, 240], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dd4acb710c139112...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), CoRoT-2130623061512.FITS → CoRoT-2130623095712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-130623.log (761d2143866b...), FinalLightCurve_CoRoT-2 b_2013-06-22.png (39eb07bc2182...), FinalLightCurve_CoRoT-2 b_2013-06-22.csv (f29bb1e78138...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 00:24Z.